Postsecondary Education in Tennessee: An ROI Analysis

By Dennis Vaughan

SUMMARY

This project seeks to analyze the return on investment (ROI) of postsecondary programs in the state of Tennessee. The goal is to explore how early career earnings differ by degree type and program of study.

Data Used:

- College Scorecard Most Recent Institution-Level Data
- College Scorecard Most Recent Data by Field of Study
- IPEDS c2024_A completions data.

Sites Referenced:

- National Center for Education Statistics: CIP Code directory
- U.S. Bureau of Labor Statistics, Occupational Employment & Wage Statistics

Technologies Used:

- Python (Pandas)
- Power BI
- Canva

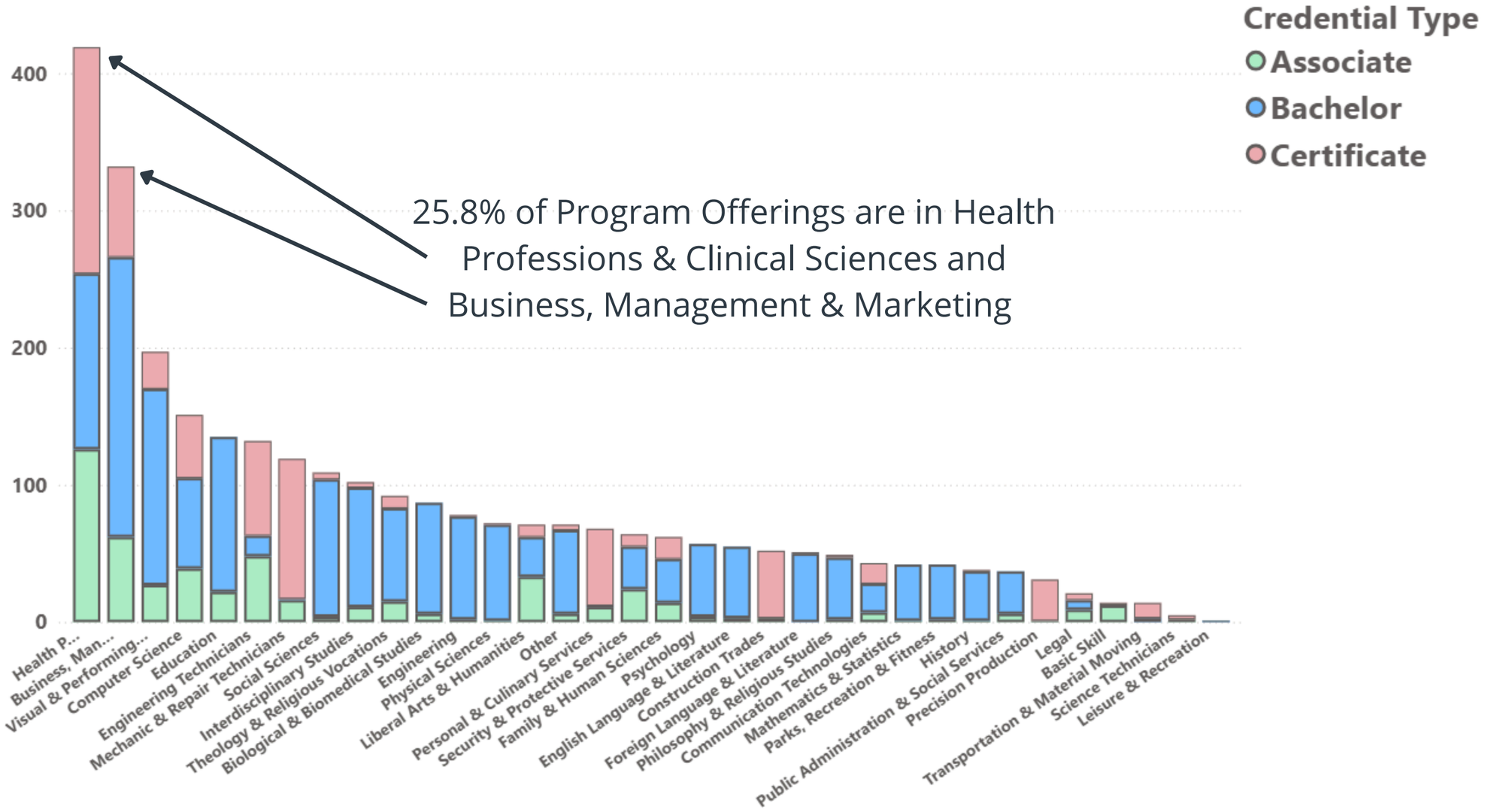
DEFINITIONS

Return on Investment (ROI):	Measure of (median earnings - median cost) / median cost
Cost Recovery:	Number of years until sum of earnings equals total program cost
CIP Family:	CIP Code group name Example: 13 - Education
CIP Code:	Classification of Instructional Program Example: 13.10 - Special Education
Earnings:	Average earnings 5 years after program completion

OVERVIEW BY CREDENTIAL LEVEL

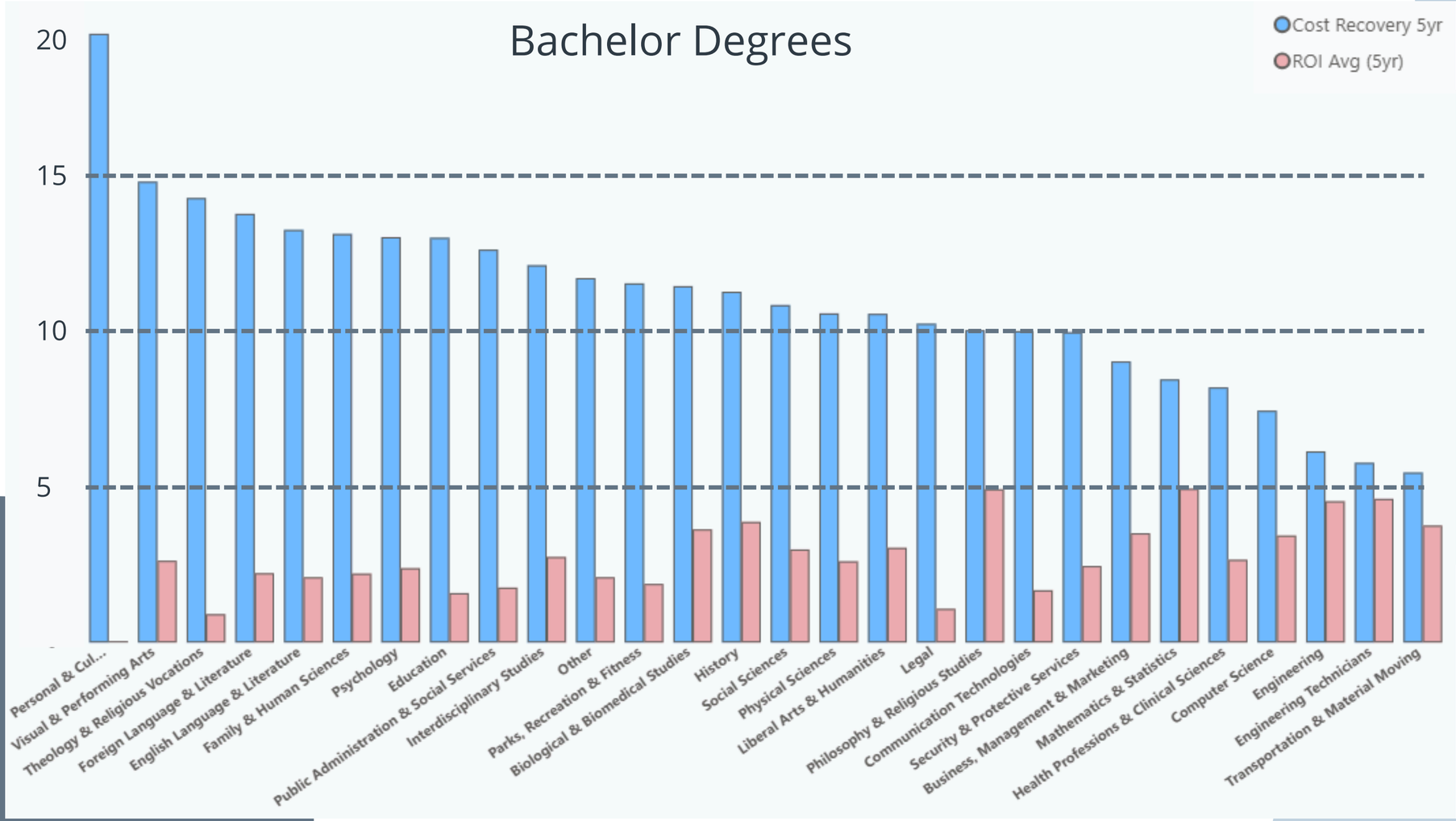
	Certificate Programs	Associate Degrees	Bachelor Degrees
Average Cost	$13,550	$30,990	$103,650
Average Earnings	$37,310	$46,660	$56, 540
Average ROI	2.99	0.90	-0.40
Recovery Years	1.82	3.32	9.17
Program Count	185	157	466

WHAT AREAS OF STUDY HAVE THE MOST PROGRAM OFFERINGS?



RECOVERY YEARS:

HOW LONG DOES IT TAKE TO EARN WHAT YOU SPENT ON YOUR DEGREE?



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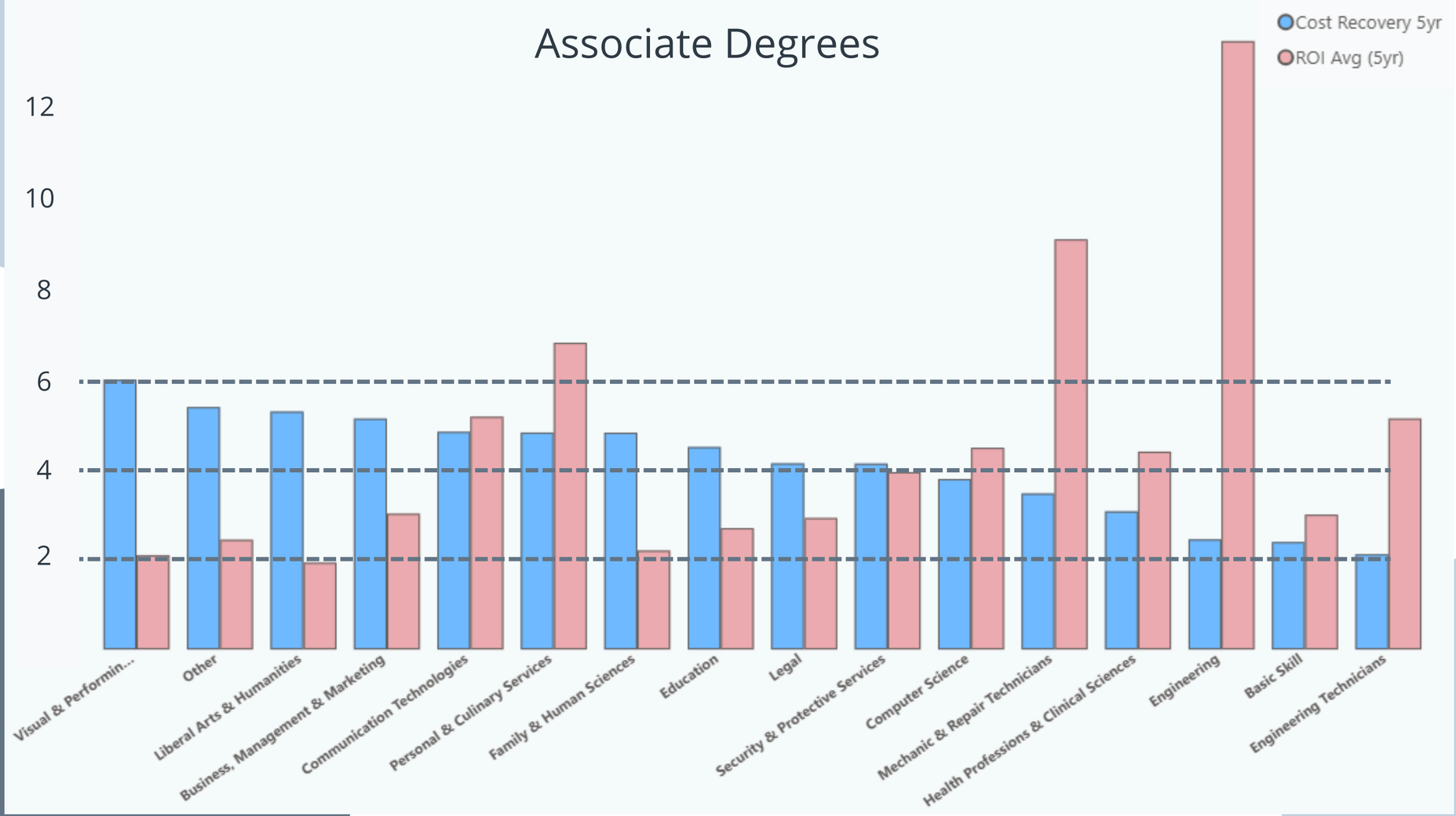

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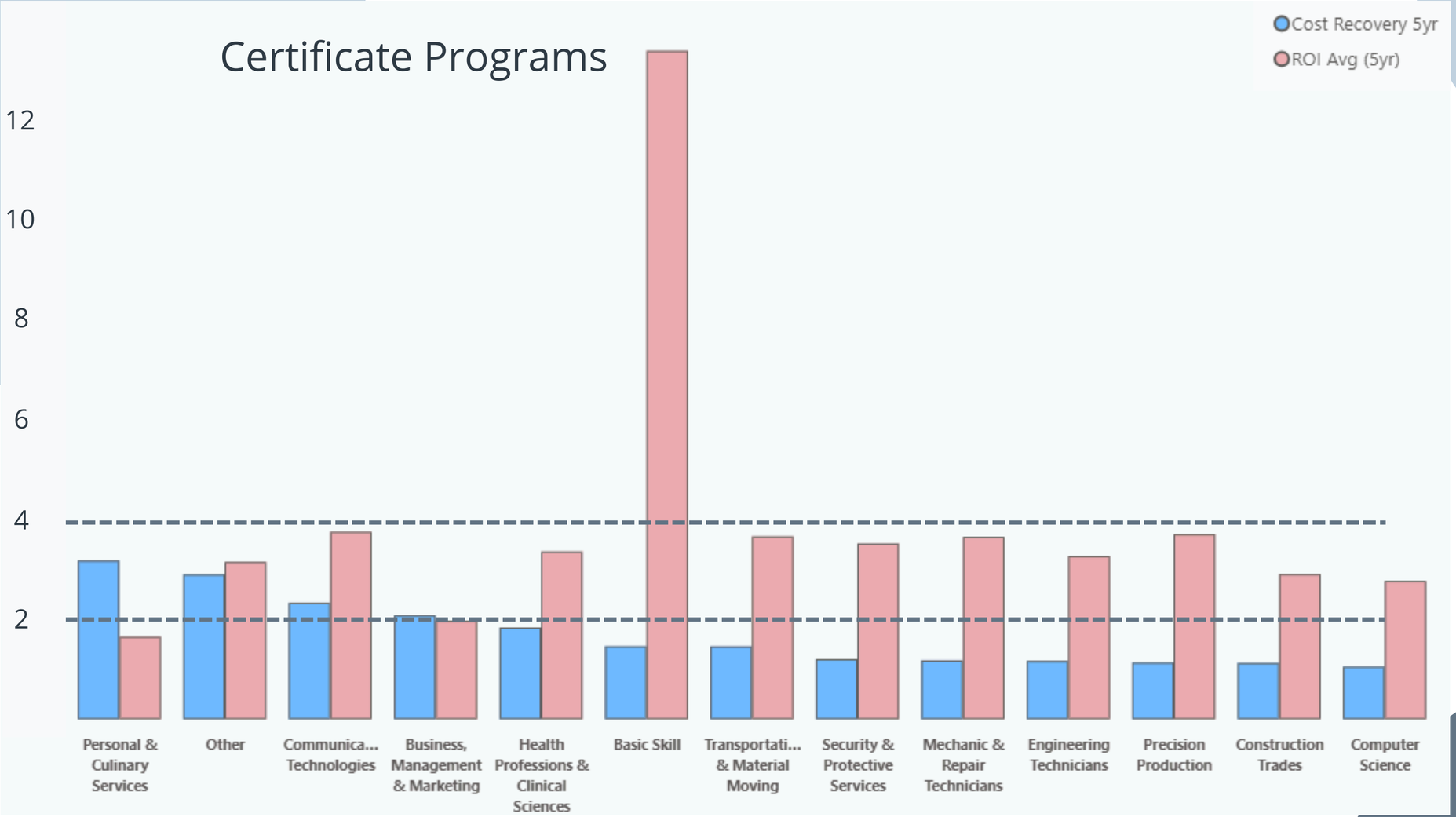
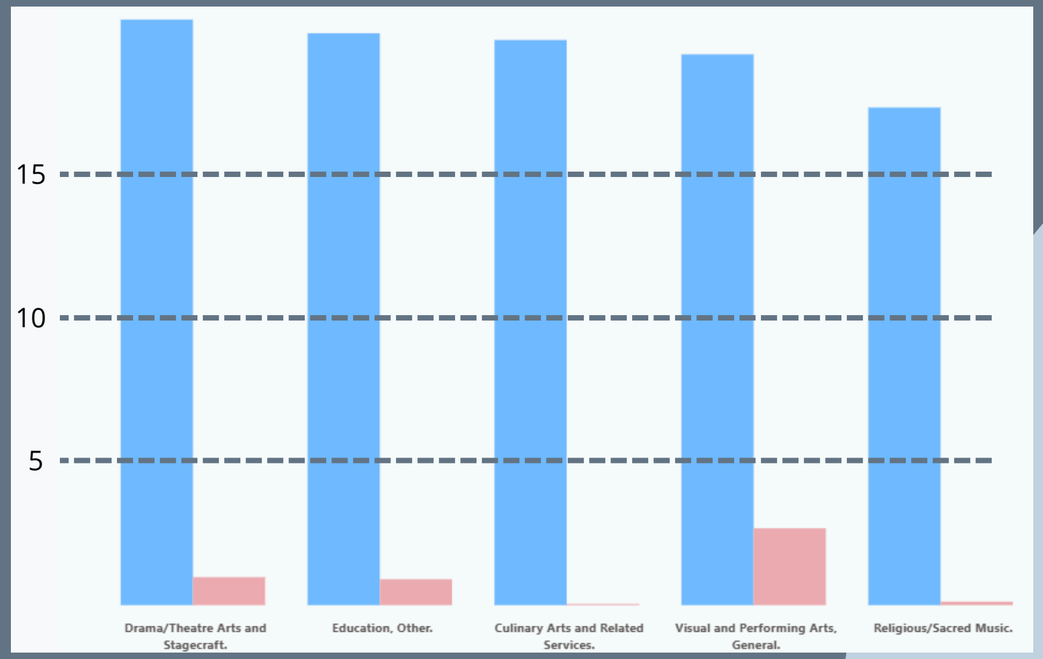


BUT ARE ALL PROGRAMS EQUAL?

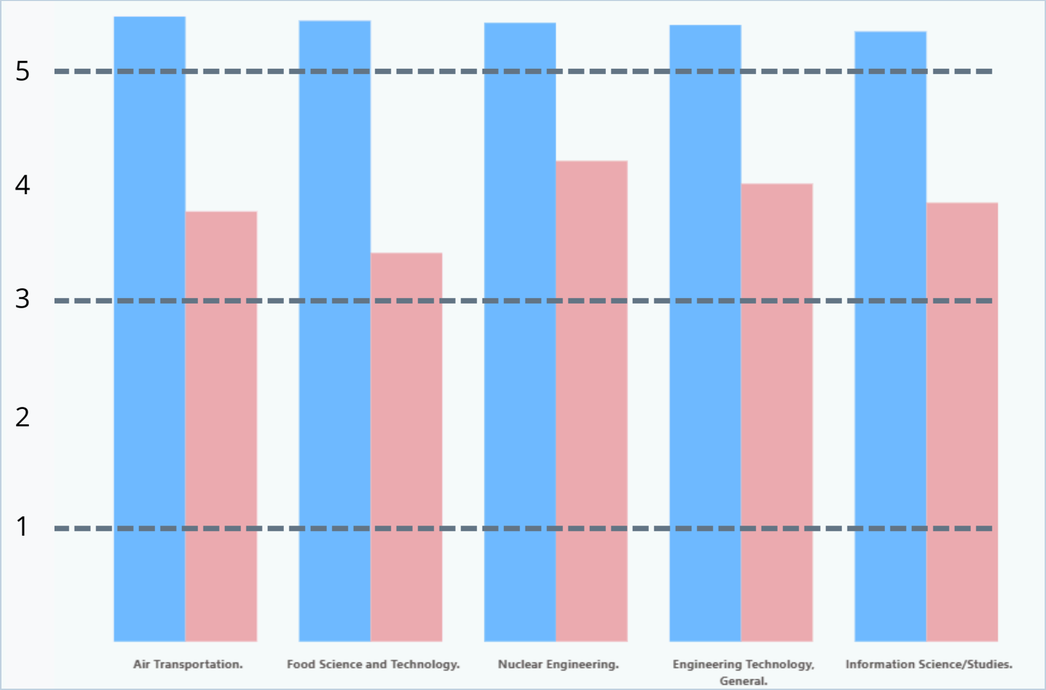
Bachelor Degrees



High Recovery Years

Cost Recovery 5yr
ROI Avg (5yr)

Low Recovery Years



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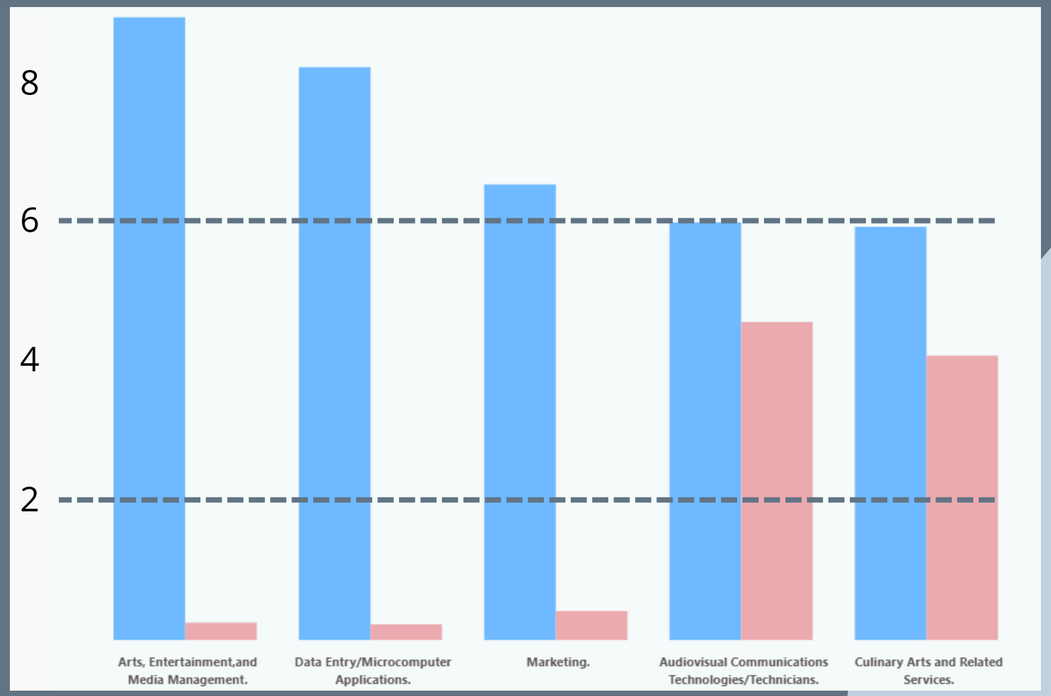
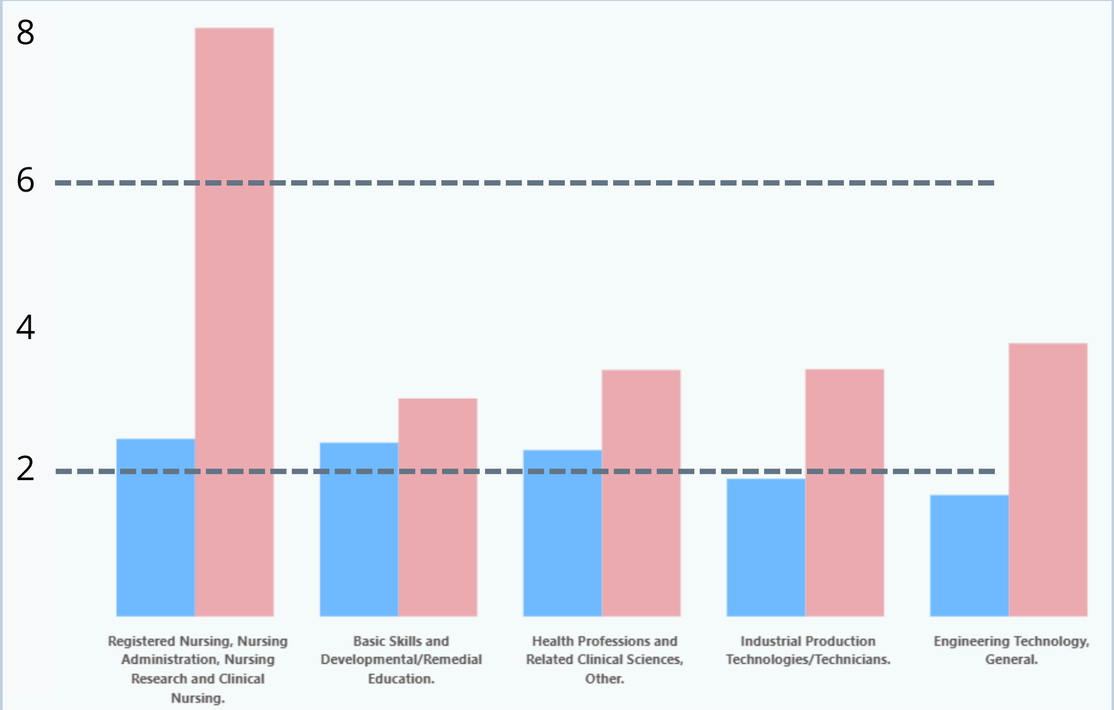
Associate Degrees



High Recovery Years

Cost Recovery 5yr
ROI Avg (5yr)

Low Recovery Years



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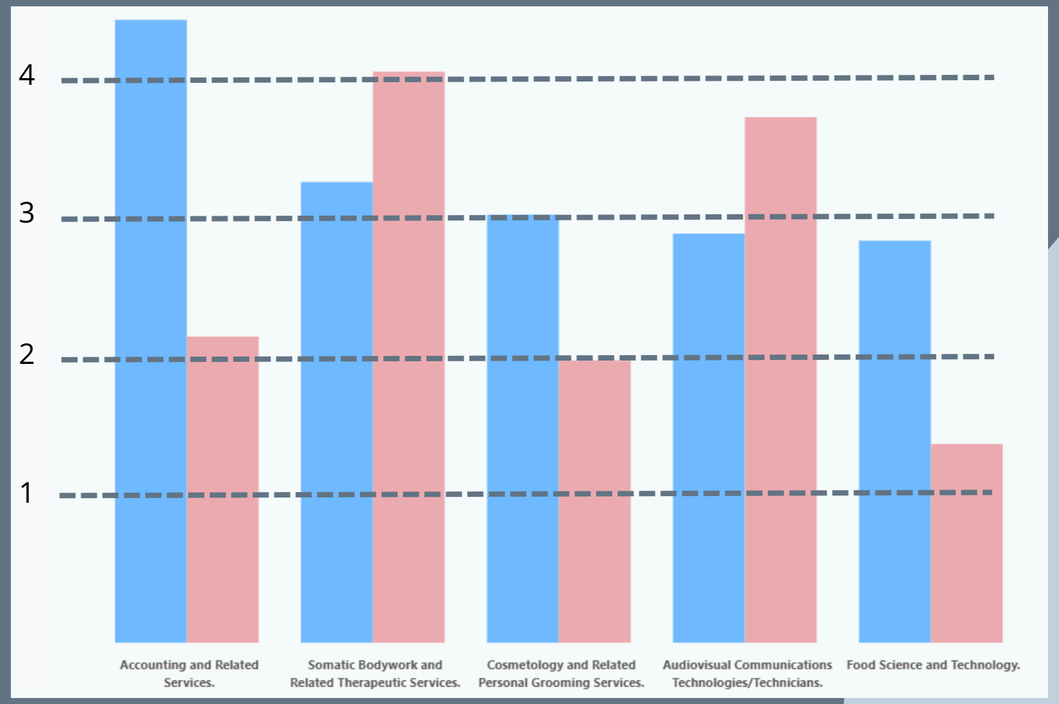
Certificate Programs



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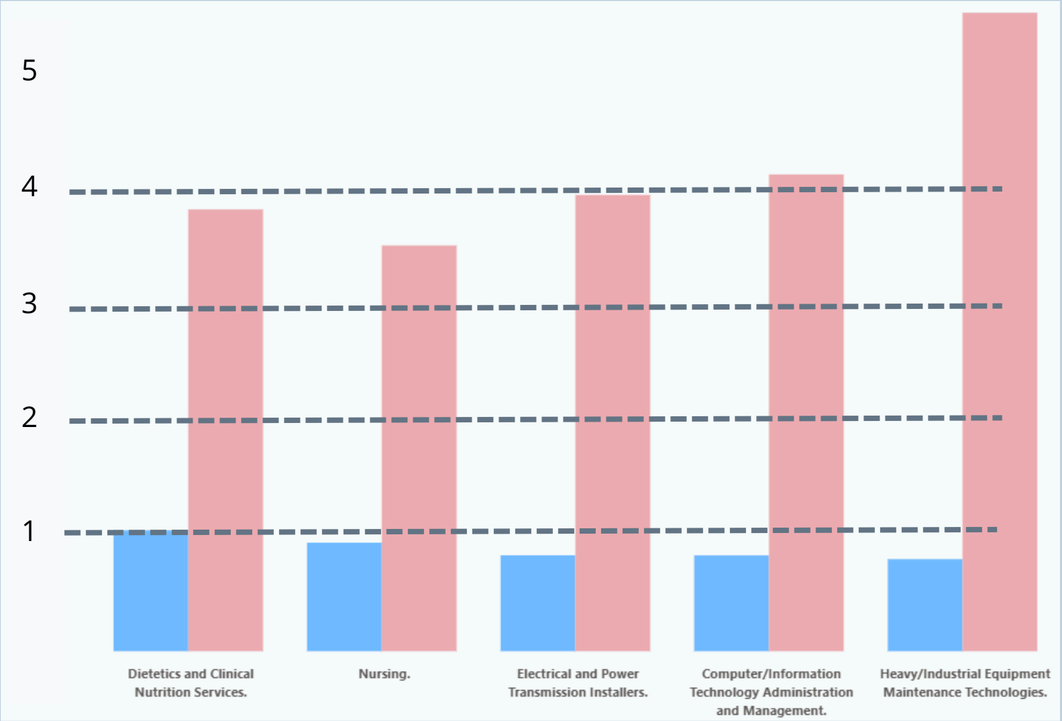

COST RECOVERY VS LONG-TERM EARNINGS POTENTIAL

Certificate

Electrician
Heavy Equipment Mechanic
IT Support Specialist
Licensed Practical Nurse

$57k - $80k

Fastest workforce entry and quickest cost recovery

Associate

Registered Nurse
Industrial Engineering Tech
Medical Laboratory Tech

$60k - $86k

Balanced pathway with stable careers and moderate recovery

Bachelor

Airline Pilot
Food Scientist
Computer Scientist
Nuclear Engineer

$82k - $140k

Highest earning potential despite longer cost recovery

Source: U.S. Bureau of Labor Statistics, Occupational Employment & Wage Statistics (2023)

CONCLUSION

- Research Question: Is a college degree still worth it?
- Short Answer: Sometimes—but outcomes vary significantly.
- Key Insight: Strong economic opportunities exist at every education level, including certificate and associate programs.
- Critical Finding: Program choice matters more than degree level. Outcomes can vary widely even within the same field of study.
- Takeaway: Education can still be a powerful investment, but students must carefully weigh cost, time, and expected return when choosing a program.

Thank you!

For follow up or additional analysis, here is how to reach me:

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